More on Hypothesis Testing

1. Null Hypothesis (H0)
2. Alternative Hypothesis (H1)
3. Significance Level (α)
4. Test Statistic
5. p- Value

P- Value

The p-value tells us how unusual our observed result would be if the null hypothesis were true.

Formula for two tailed Z- test

$$p = 2[1-\phi(|Z|)$$

Z = calcuated Z statistic.
ϕ = Cumulative Standard normal distribution function.

Example:

Z = 2

For two tailed test:

p≈0.0455

Our significance interval is:	left tailed value	right tailed
α = 0.05	0.9772	0.0228
	two tailed	0.0456

Since:

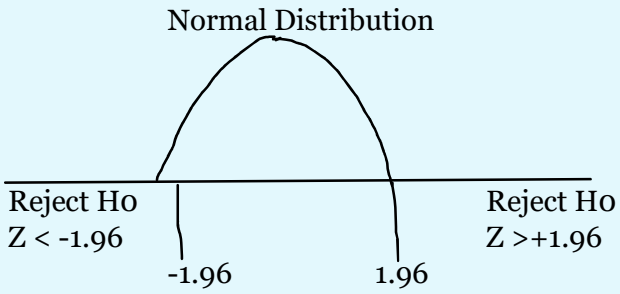
0.0455<0.05 =2*(1-NORM.S.DIST(ABS(2),TRUE))

Reject Ho 0.0455

There is statistically significant evidence against the null hypothesis at the 5% significance level.

P value decision rule

Condition	Decision
p≤α	Reject Ho
p>α	Fail to reject Ho



Does the p-value 0.03 mean there is 3% probability that null hypothesis is true?

No, It means that, the null hypothesis and statistical model, the probability of obtaining a result at least as extreme as the observed one is 3 %.

P- Value vs Critical value

Calcualted Z = 2
siginficance level (∝) =0.05
Critical values = -1.96 and +1.96

Critical Value approach:

|Z| > 1.96
2 >1.96

Decision: reject H0

P - value Approach	Critical Value Approach
0.0455 <0.05	2 > 1.96
Reject H0	Reject H0

Steps in Hypothesis testing

Delivery time

An e-commerce company claims that its average delivery time is 30 minutes.
A sample of 36 deliveries produces
Sample Mean = 32 minutes
Population Standard Deviation = 6 minutes
Significance level = 0.05

Step 1: Define the problem

Determine whether the average delivery time differs from 30 minutes.

Step 2: Formulate the Hypothesis

H0: μ = 30
H1: μ ≠ 30

This is two tailed test

Step 3: Choose the significance level

∝ = 0.05

Step 4: Select the test

Use one sample Z-test for mean because the population standard deviation is assumed known.

Step 5: Calculate the testr statistic

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

32-30
6/SQRT(36)

2

Z = 2

Step: 6 Calculate the p- value

$p \approx 0.0455$

0.0455

Step 7: Make the decision

$0.0455 < 0.05$

Reject H0.

Step 8: Interpret the result

The sample provides statistically significant evidence that the poluation's average delivery time differs from 30 minutes.

Business implication: Investigate the delivery operations to determine why the average differs from the target.
The test does not, by itself establish the cause.

Business Interpretaion

Example 1: Marketing

A company tests whther a new campign increases conversion rate:
H0: No increase in conversion rate
H1: Conversion rate increases.
Result: $p = 0.02$, $\propto = 0.05$.
Decision: Reject H0
Interpretation: The data provides statistically sgnificant evidence of an increase, assuming the test assumptions hold
Before launching widely, check the size of improvement and implementation costs.

Example: 2 Manufacturing

A factory tests whther its defect rate has decreased
H0: Defect rate has not decreased
H1: Defect rate has decreased.
Result: $p = 0.12$, $\propto = 0.05$
Decision: Fail to reject

Interpretation: The data does not provide sufficient evidence of a decrease. This does not prove that the defect rate is unchanged.

Example: 3 E-commerce

A website redesign produces a statistically significant conversion rate improvement, but the increase is very small.

Question: Should the company implement the redesign?
Not automatically, consider:
1. Size of the improvemt
2. Additional revenue
3. Development cost
4. Customer Experience.
5. Business Risk.

Key Distinction: Statistical Significance is not the same as practical significance

What are the main steps in hypothesis testing

Does rejecting Ho prove the alternative hypothesis is true?

No, It indicates that the data provides sufficeient evidence against Ho, under the chosen testing procedure.